Predicting 2024 Presidential Election Results

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CIS 1051 FINAL PROJECT

SPRING 2026

Project goals:

- Using a dataset acquired from the 2025 National Public Opinion Research Survey from the Pew Research Center, predict voting behavior using a survey of +5000 respondents and +60 survey questions.
- Find meaningful connections between survey results and the question of interest: "Who did you vote for in the 2024 Presidential Election?

First Steps

- This project started with an extensive search of political databases on the internet. Some databases were from years in the past, too small, did not have relevant political questions, were not conducted by a reputable source, were not in an Excel csv file, or were blocked behind a paywall.
- Eventually, I found a database that was large, relevant, from a reputable source, free, and in a file that I could use to make meaningful comparisons.
- Having found a usable database, it is time to execute the goal of this project.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	RESPID	MODE	LANGUAG	LANGUAG	STRATUM	INTERVIEW	INTERVIEW	ECON1MO	ECON1BM	COMTYPE2	UNITY	CRIMESAF	GOVPROT	MOREGUN	FIN_SIT	VET1	VOL12_CP	EMINUSE	INTMOB	INTFREQ	INTFREQ_(	HOME4NW	BBHOME
2	1470	2	1		10	########	########	4	2	3	2	2	2	1	3	4	2	1	1	2	2	1	2
3	2374	2	1		7	5/1/2025	5/1/2025	3	2	1	2	3	1	3	3	4	2	1	1	2	2	1	2
4	1177	3	1	10	5	3/4/2025	3/4/2025	2	1	3	2	2	1	2	3	4	2	1	1	2	2	1	2
5	15459	2	1		10	5/5/2025	5/5/2025	3	3	3	2	2	2	3	4	4	2	1	1	2	2	2	
6	9849	1	1	9	9	########	########	2	1	2	2	3	1	2	1	4	1	1	1	2	2	1	2
7	8178	3	1	9	10	########	########	2	2	1	1	2	1	2	1	4	1	2	2	6	4		
8	3682	1	1	9	4	########	########	3	2	1	2	3	2	1	2	2	1	1	1	2	2	1	2
9	6999	2	1		10	########	########	3	3	3	2	3	1	3	2	4	1	1	1	3	3	1	2
10	9945	2	1		10	5/9/2025	5/9/2025	3	2	3	2	3	1	2	3	4	2	1	1	1	1	1	2
11	1901	1	1	9	10	3/1/2025	3/1/2025	1	1	2	2	2	2	2	2	4	2	1	1	2	2	1	2
12	18470	2	1		10	5/8/2025	5/8/2025	2	3	3	2	3	2	3	2	2	1	1	1	1	1	1	2
13	18634	2	1		10	5/2/2025	5/2/2025	2	3	1	1	2	2	3	2	4	1	1	1	2	2	1	2
14	16757	2	1		9	########	########	4	3	1	2	2	2	2	4	4	1	1	1	2	2	1	2
15	9783	1	1	9	10	########	########	4	2	3	2	2	2	3	3	4	1	1	1	1	1	1	2
16	6862	2	1		9	########	########	2	1	2	2	2	2	3	3	4	2	1	1	2	2	1	2
17	11555	2	1		7	5/2/2025	5/2/2025	4	2	1	2	2	2	1	1	4	1	1	1	1	1	1	2
18	13673	1	1	9	10	3/2/2025	3/2/2025	2	3	2	1	2	2	1	2	4	2	1	1	1	1	1	2
19	14364	1	1	9	10	########	########	3	3	1	2	4	2	3	2	1	2	1	1	2	2	1	2
20	15793	3	2	9	10	3/6/2025	3/6/2025	3	2	1	1	4	2	1	3	4	2	1	1	2	2	1	4
21	18408	2	1		10	########	########	2	2	2	2	3	2	1	3	2	1	1	1	2	2	1	2
22	5945	1	1	9	10	########	########	2	3	2	2	2	2	1	1	4	1	1	1	2	2	1	2
23	16607	2	1		7	5/2/2025	5/2/2025	2	1	2	2	2	2	2	4	4	2	2	2	6	4		
24	10500	1	1	9	2	3/3/2025	3/3/2025	2	1	1	2	2	2	1	1	4	1	1	1	2	2	1	2
25	16442	1	1	9	10	########	########	2	3	2	2	2	2	3	1	4	2	1	1	1	1	1	2
26	13114	1	1	9	10	########	########	2	3	2	2	2	2	1	2	4	1	1	1	2	2	1	2

NPORS_2025_for_public_release_F

Initial Comparisons

CORREL													
ECON1MOD	ECON1BMOD	COMTYPE2	UNITY	CRIMESAFE	GOVPROTCT	MOREGUNIMPACT	FIN_SIT	VET1	VOL12_CPS	EMINUSE	INTMOB	INTFREQ	INTFREQ_COLLAPSED
0.032024776	-0.003161671	0.03326822	0.03326822	0.039899192	0.091853627	0.077213418	0.03670712	-0.009590019	0.014697105	0.007364096	0.025938659	0.017125463	0.015207933

ECON 1 B MOD	Column Labels				
Row Labels	1	2	3	99	Grand Total
1	63.62%	7.89%	17.19%	32.96%	30.64%
2	8.60%	64.36%	42.19%	29.05%	33.69%
3	27.66%	27.44%	40.63%	37.99%	35.62%
99	0.12%	0.31%	0.00%	0.00%	0.06%
Grand Total	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**

CRIMESAFE	Column Labels				
Row Labels	1	2	3	99	Grand Total
1	9.86%	9.13%	6.77%	5.59%	6.49%
2	46.66%	42.86%	34.38%	32.96%	35.56%
3	36.38%	40.43%	48.96%	50.28%	47.68%
4	5.23%	6.09%	8.33%	8.94%	8.20%
5	1.56%	1.24%	1.56%	1.12%	1.18%
99	0.30%	0.26%	0.00%	1.12%	0.89%
Grand Total	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**

PARTY SUM	Column Labels				
Row Labels	1	2	3	99	Grand Total
1	94.95%	4.59%	43.23%	36.31%	35.41%
2	3.61%	93.45%	38.54%	31.84%	40.06%
9	1.44%	1.96%	18.23%	31.84%	24.53%
Grand Total	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**

Selected Survey Responses

- Now that we have an intentional and meaningful way to select survey responses to be used in machine learning models, let us select survey features.
- Using PivotTables that show highly variable survey responses between the two candidates, I have selected the following 13 survey features

```
# Filter of chosen survey features that are highly variable and therefore more
# likely to predict voter behavior

survey_response_features = ['ECON1BMOD', 'COMTYPE2', 'UNITY', 'GOVPROTCT', 'MOREGUNIMPACT', 'RELIG', 'BORN', 'RELIMP',
                            'PRAY', 'EDUCCAT', 'RACETHN', 'GENDER', 'METRO']
```

Machine Learning Models

- I have selected three machine learning models to test the accuracy of my prediction:
  - Support Vector Machine, K Nearest Neighbor, and Random Forest
- SVM works to try to find the hyperplane that best separates two classes by maximizing the margin between the represented data points by a linear equation. This margin is the distance from the hyperplane to the nearest support vectors on each side.
- KNN works by finding the closest data points to an input and makes predictions based on the average value of the data points. KNN assigns categories based on the majority of nearby points, working by proximity and majority to make predictions.
- RF works by creating decision trees to make predictions. Each tree analyzes random parts of the data and combines them together for regression and predictions. The final prediction is the majority vote of each tree

Support Vector Machine Results

```
Support Vector Machine (SVM) Vote Prediction Results:
SVM Accuracy: 0.7631822386679001
Classification Report:
              precision    recall  f1-score   support

         1.0       0.76      0.73      0.74       505
         2.0       0.77      0.80      0.78       576

    accuracy                           0.76      1081
   macro avg       0.76      0.76      0.76      1081
weighted avg       0.76      0.76      0.76      1081

Confusion Matrix:
 [[367 138]
 [118 458]]
```

K-Nearest Neighbor Results

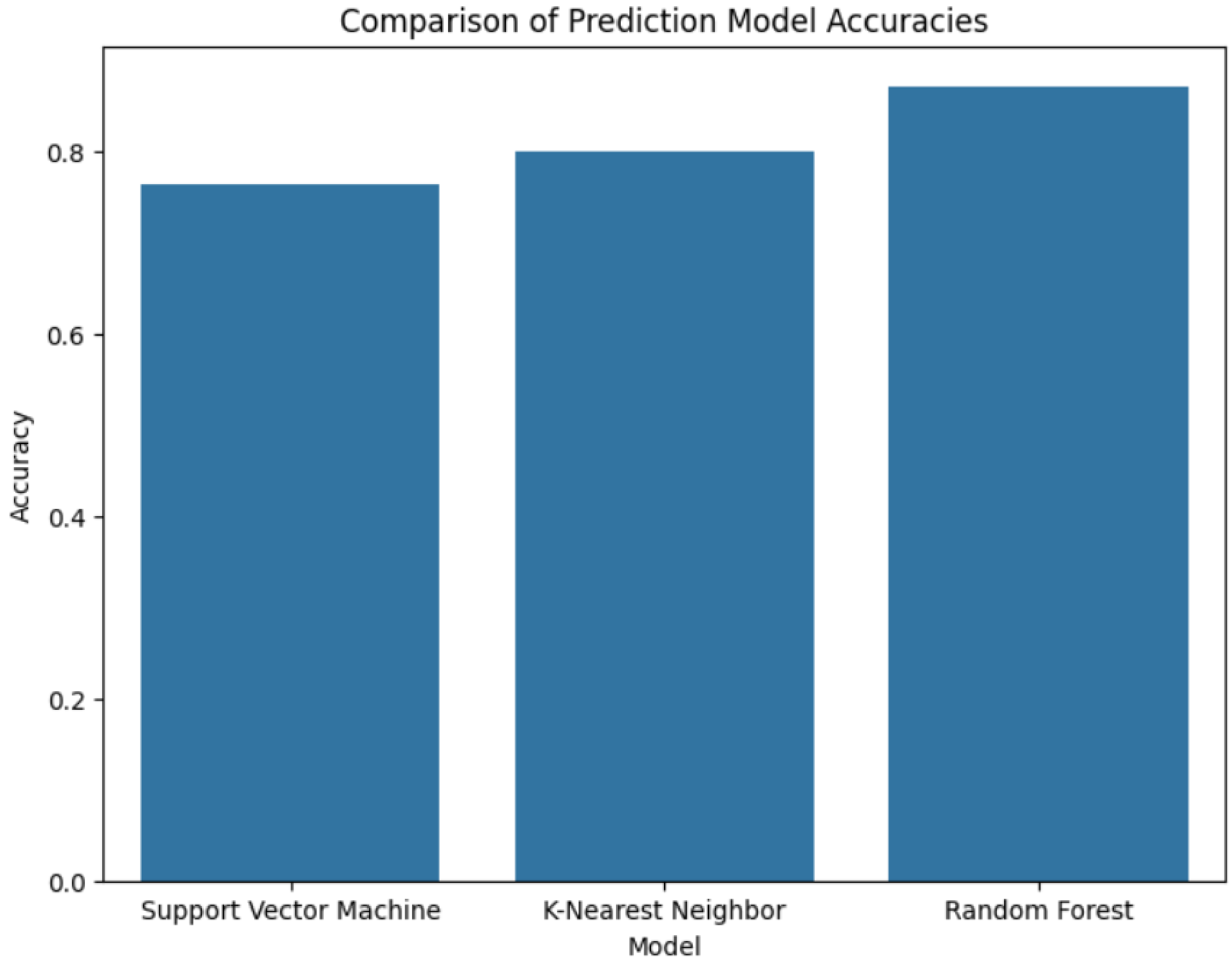
```
K-Nearest Neighbors (KNN) Results:
KNN Accuracy: 0.8001850138760407
Classification Report:
              precision    recall  f1-score   support

         1.0       0.81      0.74      0.78       505
         2.0       0.79      0.85      0.82       576

    accuracy                           0.80      1081
   macro avg       0.80      0.80      0.80      1081
weighted avg       0.80      0.80      0.80      1081

Confusion Matrix:
 [[376 129]
 [ 87 489]]
```

Random Forest Results

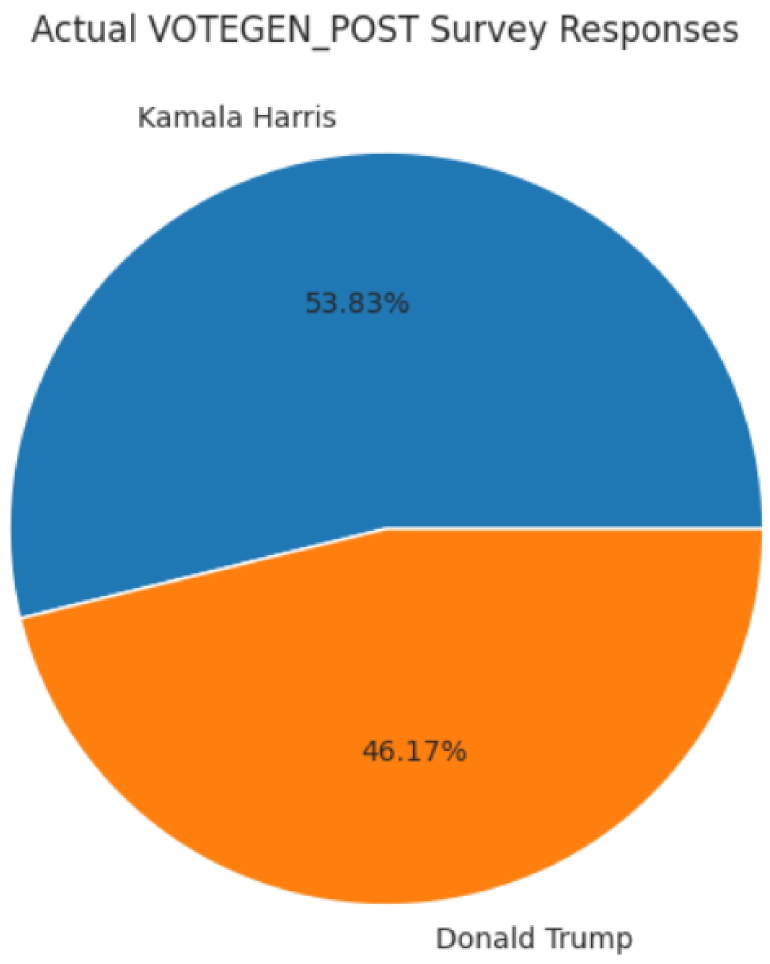
```
Random Forest (RF) Results:
Random Forest Accuracy: 0.8714153561517114
Classification Report:
              precision    recall  f1-score   support

         1.0       0.87      0.85      0.86       505
         2.0       0.87      0.89      0.88       576

    accuracy                           0.87      1081
   macro avg       0.87      0.87      0.87      1081
weighted avg       0.87      0.87      0.87      1081

Confusion Matrix:
 [[429  76]
 [ 63 513]]
```

Data visualization of machine learning models accuracy

- Bar graph that shows the accuracies between each of the machine learning models used in this project

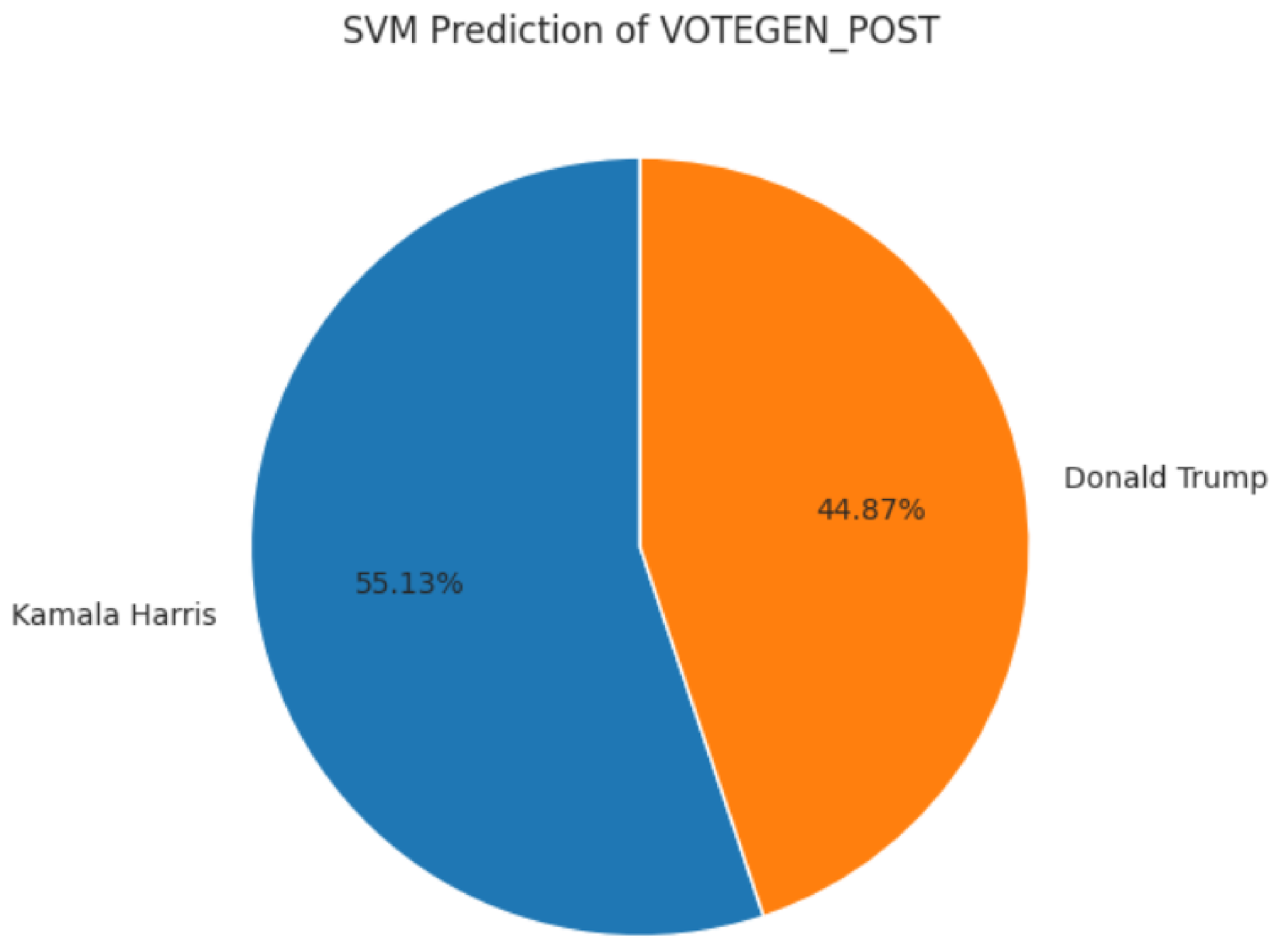


Pie Chart of actual VOTEGEN_POST results



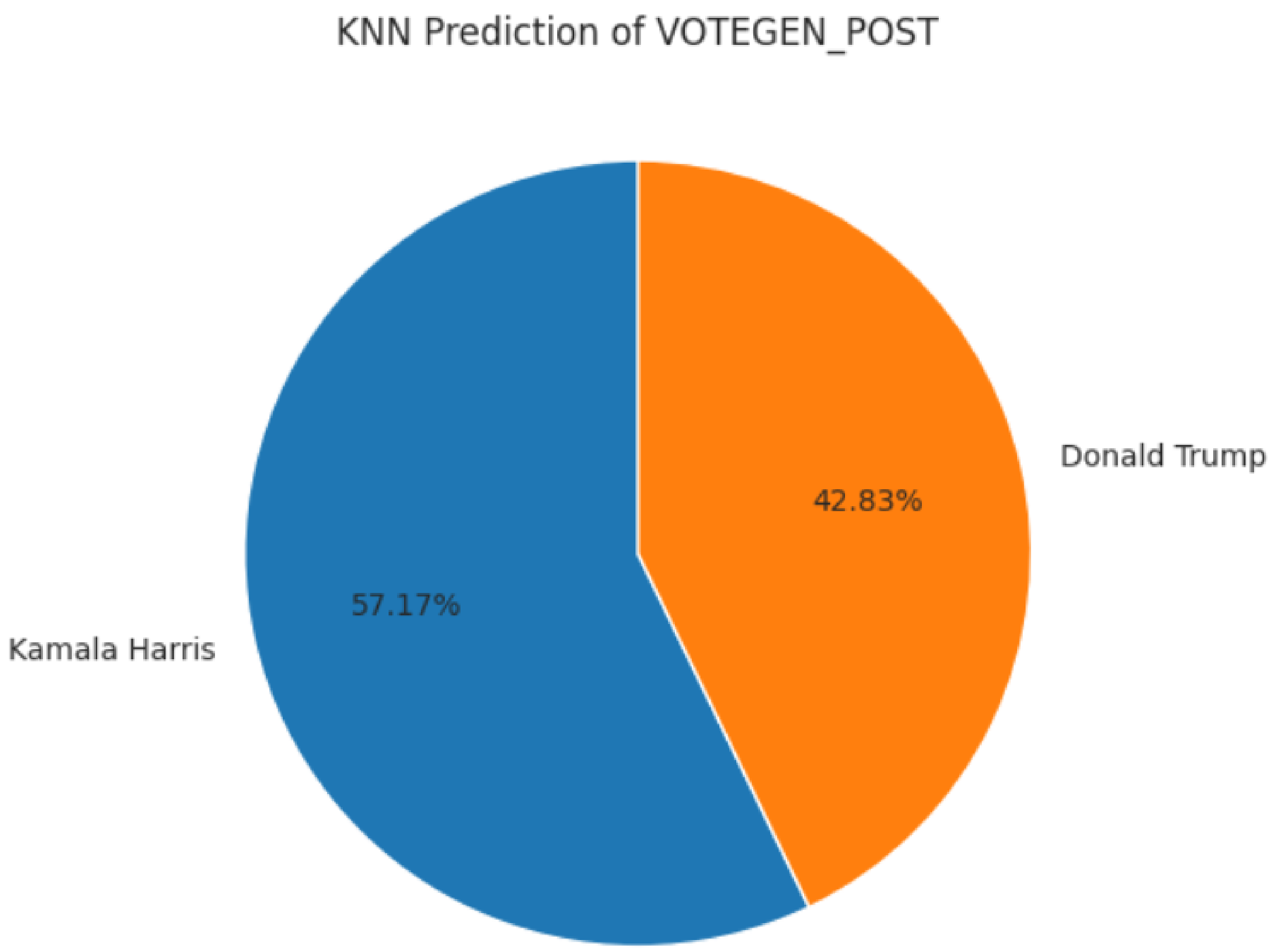
Pie Chart of SVM model prediction of VOTEGEN_POST

- +/- 1.58% difference from the actual alue



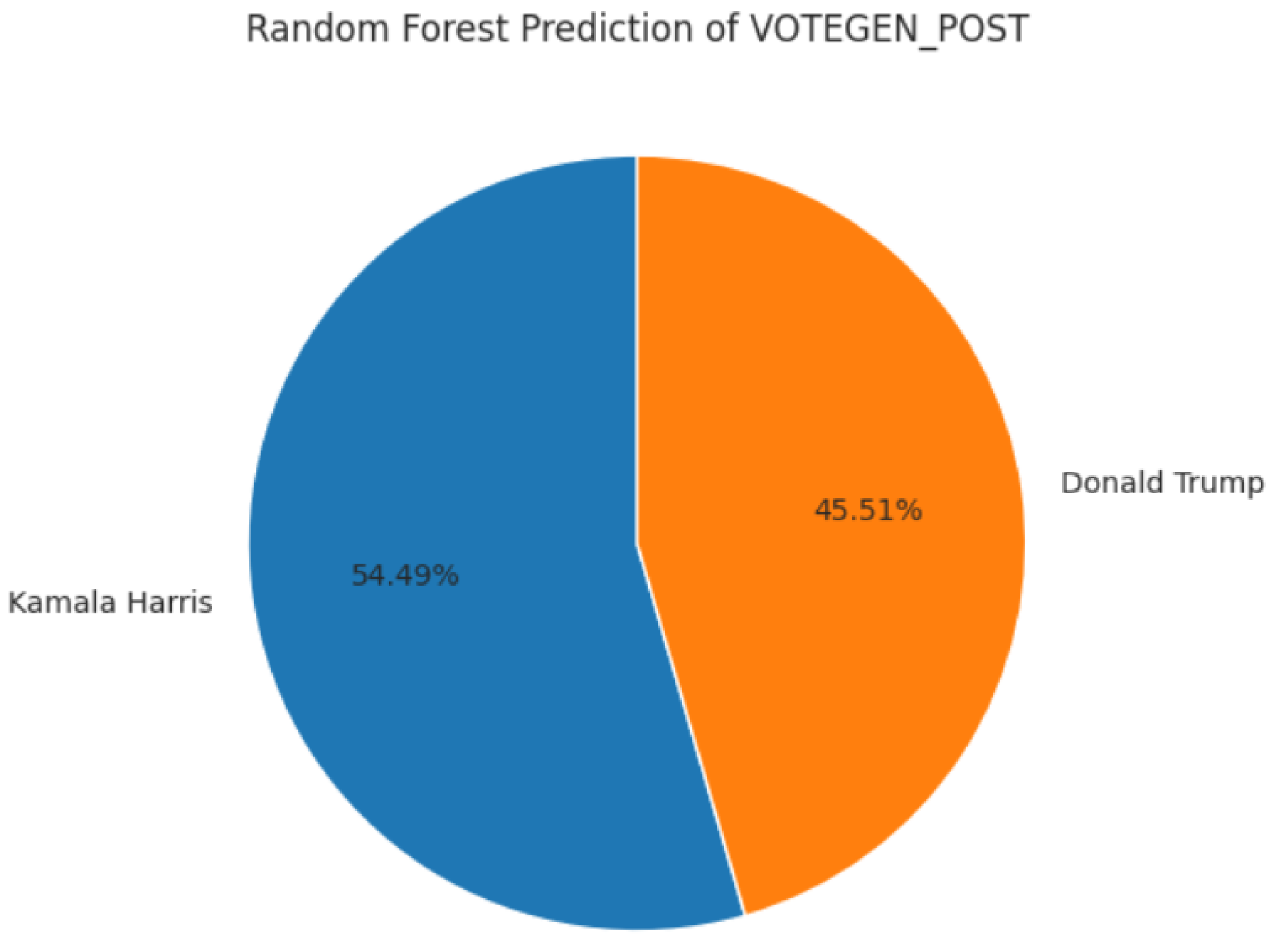
Pie Chart of KNN model prediction of VOTEGEN_POST

- +/- 4.08% difference from the actual results.



Pie chart of RF model prediction of VOTEGEN_POST

- +/- 0.84% difference from the actual results



How can this project be useful in the real world? I

- This project can be useful to a political campaign by predicting voter behavior, therefore saving time and money when they attempt to canvass or contact voters in a meaningful way.
- For example, the Harris campaign would be wasting time trying to convince a Trump voter because they are firmly set in their beliefs and canvassers could talk to someone else who is more worthwhile.
- Inversely, the Harris campaign would like to talk to a Harris voter to get-out-the-vote and potentially recruit volunteers.
- Finally, the swing voters in the middle ground are the best targets for political campaigns